INMO(26th) – 2011

Time – 3 Hrs

Instructions:

- Calculators (in any form) and protractors are not allowed.
- Ruler and compass allowed.
- Answer all the questions.

1. Let D, E, F be points on the sides of BC, CA, AB respectively of a triangle ABC such that $BD = CE = AF$ and $\angle BDF = \angle CED = \angle AFE$. Prove that ABC is equilateral.
2. Call a natural number n faithful, if there exist natural numbers $a < b < c$ such that a divides b, b divides c and $n = a + b + c$.
   (a) Show that all but a finite number of natural numbers are faithful.
   (b) Find the sum of all natural numbers which are not faithful.
3. Consider two polynomials $P(x) = a_n x^n + a_{n-1}x^{n-1} + \dots\dots\dots\dots + a_1 x + a_0$ and $Q(x) = b_n x^n + b_{n-1}x^{n-1} + \dots\dots\dots\dots + b_1 x + b_0$ with integer coefficients such that $a_n - b_n$ is a prime, $a_{n-1} = b_{n-1}$ and $a_n b_0 - a_0 b_n \neq 0$. Suppose there exist a rational number r such that $P(r) = Q(r) = 0$. Prove that r is an integer.
4. Suppose five of the nine vertices of a regular nine-sided polygon are arbitrarily chosen. Show that one can select four among these five such that they are the vertices of a trapezium.
5. Let ABCD be a quadrilateral inscribed in a circle Γ. Let E, F, G, H be the mid points of the arcs AB, BC, CD, and DA of the circle Γ. Suppose $AC.BD = EG.FH$. Prove that AC, BD, EG, FH are concurrent.
6. Find all functions $f : \Re \to \Re$ such that $f(x+y)f(x-y) = (f(x) + f(y))^2 - 4x^2 f(y)$ for all $x, y \in \Re$, where $\Re$ denotes the set of all real numbers.

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